

No analytic renorming for uncountable Lorentz preduals

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Abstract

Bible asks in Remark 3.10 of [1] whether the uncountable Lorentz preduals $d_*(w, 1, A)$, already shown there to admit C^∞ -smooth renormings, admit analytic renormings. We give a candidate full negative answer. For every standard decreasing Lorentz weight w and every uncountable set A , the Banach space $d_*(w, 1, A)$ admits no equivalent analytic norm. The proof is stronger: no continuous polynomial separates the origin from the unit sphere. The finite- ℓ_q cases, where diagonal high-degree polynomials exist, are handled by normalized flat averages.

1 Source question

The source is V. Bible, *Using boundaries to find smooth norms*, arXiv:1311.1408. The relevant passage is Example 3.9 and Remark 3.10 on page 15:

USING BOUNDARIES TO FIND SMOOTH NORMS

15

Example 3.9. $X = d_*(w, 1, A)$ has a C^∞ smooth equivalent renorming that ε -approximates the original norm.

Proof. Let $y \in S_X$. Since $\bar{y} \in c_0$, there exists $k \in \mathbb{N}$ such that $\bar{y}^k = 1$. It can also be shown $y \in c_0(A)$ and thus the supremum in the definition of $\bar{y}^k$ is attained. Following this, there exists $a_0, a_1, \dots, a_{k-1} \in A$ such that

$$1 = \bar{y}^k = \frac{\sum_{i=0}^{k-1} |y(a_i)|}{\sum_{i=0}^{k-1} w_i}.$$

Setting $\sigma = \{a_0, a_1, \dots, a_{k-1}\}$, we have $\|P_\sigma(y)\| = 1$. By Corollary 3.6, X has a C^∞ smooth equivalent renorming that ε -approximates the original norm. □

Remark 3.10. The space $X = d_*(w, 1, A)$ for A uncountable is a new example of a space with a C^∞ smooth renorming. It is not yet known if X has an analytic renorming.

Remark 3.11. In Theorem 2.1 and Corollary 3.1 we cannot drop the

Remark 3.10 says that, for uncountable A , $X = d_*(w, 1, A)$ is a new example of a space with a C^∞ -smooth renorming and that it is not yet known whether X has an analytic renorming.

2 Definitions

Let $w = (w_j)_{j \geq 1}$ be a standard Lorentz weight: $w_1 = 1$,

$$w_1 \geq w_2 \geq \cdots > 0, \quad w_j \rightarrow 0, \quad \sum_{j=1}^{\infty} w_j = \infty.$$

Put $W_m = \sum_{j=1}^m w_j$. For a set A , $d_*(w, 1, A)$ is the space of all scalar functions x on A such that

$$\bar{x}_m = \sup \left\{ \frac{\sum_{i=1}^m |x(a_i)|}{W_m} : a_1, \dots, a_m \in A \text{ distinct} \right\}$$

defines a null sequence. The norm is

$$\|x\|_* = \sup_m \bar{x}_m.$$

Equivalently, if (x_j^*) is the decreasing rearrangement of the nonzero values of $|x|$, then

$$\|x\|_* = \sup_m \frac{\sum_{j=1}^m x_j^*}{W_m}.$$

3 Idea of the proof

The previous partial packet covered the case $w \notin \ell_q$ for every finite q : then every continuous homogeneous polynomial is countably supported, and a fresh coordinate defeats analytic separation.

The missing finite- ℓ_q case is subtler because diagonal polynomials such as

$$x \mapsto \sum_{a \in A} x(a)^q$$

may be continuous. The replacement obstruction is not support, but flattening. For a finite set $F \subset A$ of size m , put

$$u_F = \frac{W_m}{m} \sum_{a \in F} e_a.$$

Then $\|u_F\|_* = 1$. The main lemma says that, after choosing F far enough outside any prescribed countable set, every fixed finite family of continuous homogeneous polynomials is arbitrarily small on u_F .

For a diagonal q -polynomial this is the elementary estimate

$$\sum_{a \in F} u_F(a)^q = \frac{W_m^q}{m^{q-1}} \longrightarrow 0 \quad (w \in \ell_q).$$

For general polynomials the same mechanism appears after a Ramsey thinning of the coordinate coefficients by equality pattern. Any nonzero limiting coefficient with block sizes $r_1, \dots, r_s$ forces $w \in \ell_{r_i}$ for every block size r_i ; then

$$m^s \left(\frac{W_m}{m} \right)^{r_1 + \cdots + r_s} = \prod_{i=1}^s \frac{W_m^{r_i}}{m^{r_i-1}} \longrightarrow 0.$$

Thus no polynomial can separate the unit sphere. An analytic norm would yield an analytic separating function; truncating its local Taylor expansion and applying the flattening lemma gives the contradiction.

4 The flattening lemma

Lemma 1. *If $r \geq 1$, $w \in \ell_r$, and $r > 1$, then*

$$\frac{W_m^r}{m^{r-1}} \longrightarrow 0.$$

For $r = 1$ this conclusion is irrelevant here, since Lorentz weights satisfy $w \notin \ell_1$.

Proof. Fix $M < m$. By Holder's inequality,

$$\sum_{j=M+1}^m w_j \leq (m-M)^{1-1/r} \left(\sum_{j>M} w_j^r \right)^{1/r}.$$

After division by $m^{1-1/r}$, the initial sum $\sum_{j \leq M} w_j$ disappears as $m \rightarrow \infty$. Taking the limsup and then letting $M \rightarrow \infty$ gives $W_m/m^{1-1/r} \rightarrow 0$, which is the claim. $\square$

Lemma 2 (flat averages kill finite polynomial families). *Let A be uncountable, $X = d_*(w, 1, A)$, $C \subset A$ countable, and $\mathcal{F}$ a finite family of continuous homogeneous scalar polynomials on X . For every $\varepsilon > 0$ there is a finite set $F \subset A \setminus C$ such that, with*

$$u_F = \frac{W_{|F|}}{|F|} \sum_{a \in F} e_a,$$

one has

$$\|u_F\|_* = 1 \quad \text{and} \quad |P(u_F)| < \varepsilon \quad (P \in \mathcal{F}).$$

Proof. First observe that $\|u_F\|_* = 1$ when $|F| = m$. Indeed, for $k \leq m$,

$$\frac{k(W_m/m)}{W_k} \leq 1$$

because $W_k/k \geq W_m/m$, while for $k \geq m$ the ratio is $W_m/W_k \leq 1$. Equality occurs at $k = m$.

We prove the polynomial estimate. Let d be the largest degree in $\mathcal{F}$, and let $\check{P}$ denote the associated symmetric multilinear form. A standard Ramsey diagonal thinning gives an infinite sequence $B = (b_j)$ in $A \setminus C$ with the following property: for every $P \in \mathcal{F}$ and every partition π of $\{1, \dots, \deg P\}$, the coefficient

$$\check{P}(e_{b_{i_1}}, \dots, e_{b_{i_{\deg P}}})$$

has a limit $L(P, \pi)$ along increasing choices of distinct indices whose equality pattern is π . This is just the usual finite Ramsey theorem applied successively to rational approximations of the finitely many bounded coefficient arrays.

Fix one such pair (P, π) . Let the block sizes of π be $r_1, \dots, r_s$, so that $r_1 + \dots + r_s = \deg P$. We record the standard consequence of continuity: if $L(P, \pi) \neq 0$, then

$$w \in \ell_{r_i} \quad (1 \leq i \leq s).$$

Indeed, if $w \notin \ell_{r_i}$ for some block size, take s disjoint subsequences of B , one for each block of π . On the r_i argument positions belonging to the i -th block, insert vectors

$$x_{i,t} = \sum_{j=1}^M \eta_{i,t,j} w_j e_{b_j^i}, \quad |\eta_{i,t,j}| = 1.$$

Each of these vectors has norm at most one. Averaging the signs separately inside each block and choosing the last sign in each block to align the diagonal product kills all mixed terms and leaves the terms with the equality pattern π . Since the corresponding coefficients are eventually within $|L(P, \pi)|/2$ of $L(P, \pi)$, continuity of $\dot{P}$ would force

$$\prod_{i=1}^s \sum_{j=1}^M w_j^{r_i}$$

to remain bounded as $M \rightarrow \infty$, impossible if one factor diverges.

Now choose m large. In the expansion of $P(u_F)$, the contribution of the pattern π is, up to an error made small by taking the block of indices far enough out in B ,

$$L(P, \pi) (m)_s \left(\frac{W_m}{m} \right)^{\deg P},$$

where $(m)_s = m(m-1) \cdots (m-s+1)$. If $L(P, \pi) = 0$, choose the block far enough out so that the coefficient error is less than ε divided by the finite combinatorial factor. If $L(P, \pi) \neq 0$, the preceding paragraph gives $w \in \ell_{r_i}$ for every i , and Lemma 1 gives

$$(m)_s \left(\frac{W_m}{m} \right)^{\deg P} \leq \prod_{i=1}^s \frac{W_m^{r_i}}{m^{r_i-1}} \rightarrow 0.$$

Since there are only finitely many polynomials and equality patterns, a single large m and a sufficiently far-out block $F \subset B$ of size m make all contributions smaller than ε . This proves the lemma. $\square$

Corollary 1. *The space $d_*(w, 1, A)$, for uncountable A , admits no separating polynomial.*

Proof. Let $Q = \sum_{j=0}^d P_j$ be a continuous polynomial, decomposed into homogeneous parts, and suppose $Q(0) = 0$. Apply Lemma 2 to the finite family $\{P_1, \dots, P_d\}$. It gives unit vectors u_F with $|Q(u_F)|$ arbitrarily small. Hence Q cannot be bounded away from zero on the unit sphere. $\square$

5 No analytic norm

Fact 1 (analytic separation with local Taylor control). *If a normed space admits an equivalent analytic norm $\|\cdot\|$, then there are $\rho > 0$ and an analytic function $f: X \rightarrow \mathbb{R}$, represented by its Taylor series at the origin on a norm ball containing $\{x: \|x\| \leq \rho\}$, such that*

$$f(0) = 0 \quad \text{and} \quad \inf_{\|x\|=\rho} f(x) > 1.$$

This is the standard separating-function consequence of an analytic norm used in [2]. The usual affine-hyperplane construction of the coercive analytic separator may be made with the translated hyperplane far enough from the origin, and then rescaled, so that the separating sphere lies inside the convergence ball of the Taylor expansion at zero.

Theorem 1. *Let A be uncountable and let w be any standard decreasing Lorentz weight. Then*

$$X = d_*(w, 1, A)$$

does not admit an equivalent analytic norm.

Proof. Assume that X admits an equivalent analytic norm $\|\cdot\|$. Choose constants $0 < a \leq b < \infty$ such that

$$a\|x\|_* \leq \|x\| \leq b\|x\|_* \quad (x \in X).$$

Let f and ρ be as in Fact 1. Then $f(x) > 1$ whenever $\|x\| = \rho$, and its Taylor series at the origin converges absolutely and uniformly on a norm ball containing $\{x : \|x\| \leq \rho\}$. Write

$$f(x) = \sum_{n=0}^{\infty} P_n(x)$$

on that ball. Since $f(0) = 0$, $P_0 = 0$. Choose N so that the tail $\sum_{n>N} P_n$ has absolute value $< 1/4$ on $\{x : \|x\| \leq \rho\}$.

Apply Lemma 2 to the finite family $\{P_1, \dots, P_N\}$, with $\varepsilon > 0$ so small that

$$\sum_{n=1}^N \left(\frac{\rho}{a}\right)^n |P_n(u_F)| < \frac{1}{4}.$$

For the resulting vector u_F , put

$$x = \frac{\rho}{\|u_F\|} u_F.$$

Then $\|x\| = \rho$, while $\|u_F\| \geq a$ because $\|u_F\|_* = 1$. Hence the homogeneous scaling estimate above gives

$$\left| \sum_{n=1}^N P_n(x) \right| < \frac{1}{4},$$

and the tail contributes $< 1/4$. Thus $|f(x)| < 1/2$, contradicting $f(x) > 1$ on the $\|\cdot\|$ -sphere of radius ρ . $\square$

6 Verification and novelty

The earlier slow-weight packet proved the special case $w \notin \ell_q$ for every finite q . This packet replaces the countable-support argument by Lemma 2, which also covers the finite- ℓ_q regime where diagonal polynomials are continuous.

The run indexes were searched for 1311.1408, `Using boundaries to find smooth norms`, `d_*(w,1,A)`, `Lorentz predual`, and `analytic renorming`. The only exact hit was the prior slow-weight partial packet, now superseded by this one. Web searches on 2026-06-20 for the exact Lorentz-predual analytic-renorming question and close phrases did not locate a later full answer. The Dantas–Hajek–Russo paper [2] supplies the analytic-separation template for $c_0(\omega_1)$, but not this Lorentz-predual flattening argument.

7 Human review recommendation

Review as a candidate full negative answer to Bible’s Remark 3.10. The main technical point is Lemma 2, especially the Ramsey thinning and the sign-averaging step that forces nonzero spreading coefficients of equality pattern $(r_1, \dots, r_s)$ to satisfy $w \in \ell_{r_i}$ for every block size. The rest of the analytic contradiction is standard once that lemma is accepted.

References

- [1] V. Bible, *Using boundaries to find smooth norms*, arXiv:1311.1408, 2013.
- [2] S. Dantas, P. Hájek, and T. Russo, *Smooth norms in dense subspaces of Banach spaces*, arXiv:1911.09611, 2019.